

Savitha Vijayarangan

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Education

M.S., Applied Data Intelligence

San José State University

2025 – Present

San Jose, CA

Key Skills: Data Warehousing, Distributed Systems, Big Data, Machine Learning, Deep Learning, Generative AI

B.Tech, Electronics & Communication Engineering

B.S. Abdur Rahman University

2013 – 2017

India

Business Analytics Certification

Harvard Business School Online – Hypothesis Testing, Regression, Statistical Inference

2023

Experience

Instructional Student Assistant – Deep Learning & Generative AI

San José State University

Jul 2026 – Present

San Jose, CA

Key Skills: Model Deployment, Distributed Training, Curriculum Design, Python

- Developed and refined technical curriculum for Deep Learning (DATA 255) and GenAI (DATA 266) graduate courses, mentoring 100+ students on backend model deployment and distributed training; automated grading pipelines and designed scalable cloud-based lab environments.

ML Engineer Intern

HexasenseAI

Jun 2025 – Aug 2025

Austin, TX (Remote)

Key Skills: PyTorch, Feature Engineering, Responsible AI, Healthcare ML

- Deployed optimized ML pipelines that reduced inference latency by 30%, and built clinical risk prediction models on healthcare data, applying Responsible AI practices to ensure fairness, transparency, and reliability.

System Consultant (Data & Analytics Engineer)

Larsen & Toubro Infotech (LTI)

Jan 2020 – May 2021

India

Key Skills: Python, SQL, Airflow, Snowflake, ETL

- Developed ETL pipelines handling 1M+ records/day using Python, SQL, and Airflow, improving data availability by 30% and integrating 5+ sources into a Snowflake warehouse; optimized workflows, cutting latency 25% and saving 10+ hrs/week.

Software Engineer

Larsen & Toubro Infotech (LTI)

Sep 2017 – Dec 2019

India

Key Skills: Python, SQL, REST APIs, Agile

- Built backend services in Python handling 50K+ transactions/day, improving system performance by 20%, and delivered 10+ production features via REST APIs in Agile sprints.
- Implemented data processing logic for large-scale datasets using SQL, achieving 99.5% data accuracy while reducing bug rates by 15% and downtime by 30%.

Projects

Wildfire Edge Sentinel: Offline Smoke Detection on Edge Hardware (Hackathon)

GitHub

Key Skills: YOLO11, Qwen2.5-VL, LoRA, ONNX, Docker, Edge AI

- Built an offline-first wildfire smoke detection pipeline on an HP ZGX Nano edge device that sends alerts, not video, to the cloud, using YOLO11 for ~40ms/frame detection and multi-frame persistence gating.
- Integrated a LoRA-tuned Qwen2.5-VL vision-language model for once-per-event severity analysis, cutting VLM calls from 266 to 11 on a 6-tower run.

Kayak Travel Simulation System (AI-Powered Booking & Chatbot)

GitHub

Key Skills: Python, FastAPI, React, Kafka, MongoDB, LangChain, OpenAI API

- Engineered a distributed 3-tier microservices architecture with Apache Kafka to orchestrate real-time travel booking feeds, and built an AI chatbot with LangChain/OpenAI API improving response accuracy by 25%.

AquaShift: Weather-Responsive Workload Optimization for Data Centers (Hackathon)

GitHub

Key Skills: Python, Guild.ai, Jua AI API, Render, Composio, Agentic Orchestration

- Built an autonomous multi-agent scheduling framework relocating AI inference workloads in real time based on a custom Water Stress Index, orchestrating sub-agents via Guild.ai to cut water usage.

Deep Learning-Based Image Super-Resolution System

Drive

Key Skills: Python, PyTorch, CNN, ONNX, NPU Deployment

- Trained a custom DNN to reconstruct high-quality images from degraded inputs; deployed via ONNX-to-MXQ on NPU.